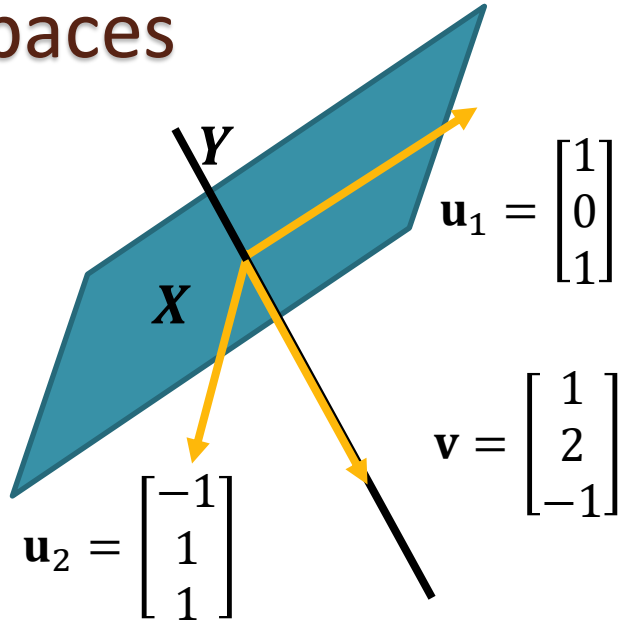
Chapter 5
Orthogonality

5.2 Orthogonal Subspaces



Motivation

- In 5.1,
  - We defined scalar product $\mathbf{x}^T\mathbf{y}$, and defined when two vectors are orthogonal:

$$\mathbf{x} \perp \mathbf{y} \Leftrightarrow \mathbf{x}^T\mathbf{y} = 0$$

  - Given $\mathbf{x}$ and a line $l = Span(\mathbf{y})$, we can write $\mathbf{x} = \mathbf{p} + \mathbf{z}$, where $\mathbf{p}$ is in l, and $\mathbf{z}$ is orthogonal to $\mathbf{p}$.
- In 5.2, we will see that
  - When two subspaces are orthogonal
  - Given any subspace S in $\mathbb{R}^n$, there is an orthogonal complement $S^{\perp}$, so that any vector $\mathbf{x}$ can be written uniquely as a vector $\mathbf{p}$ in S, plus a vector $\mathbf{z}$ orthogonal to S.
  - Given an $m \times n$ matrix A, there will be 2 important orthogonal subspaces in $\mathbb{R}^n$, and 2 important orthogonal subspaces in $\mathbb{R}^m$.

Orthogonal subspaces

Definition

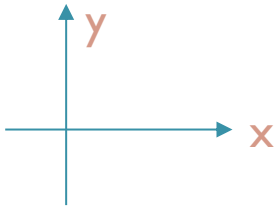
Two subspaces X and Y of $\mathbb{R}^n$ are said to be ***orthogonal* 正交** if $\mathbf{x}^T\mathbf{y}=0$ for every $\mathbf{x} \in X$ and every $\mathbf{y} \in Y$.

If X and Y are orthogonal, we write $X \perp Y$.

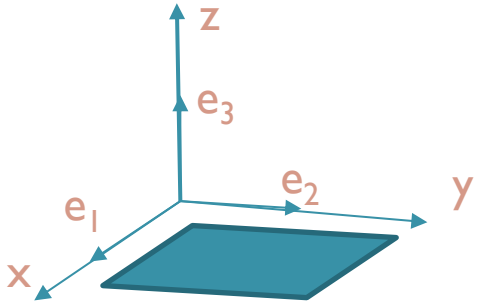
***Note**: X and Y are subspaces of the same vector space $\mathbb{R}^n$.*

Example:

- In $\mathbb{R}^2, Span(\boldsymbol{e}_1) \perp Span(\boldsymbol{e}_2)$.



- In $\mathbb{R}^3, Span(\boldsymbol{e}_1) \perp Span(\boldsymbol{e}_2)$,
  $Span(\boldsymbol{e}_1, \boldsymbol{e}_2) \perp Span(\boldsymbol{e}_3)$.



Example: Orthogonal subspace

In $\mathbb{R}^4$, show that $X = Span(\mathbf{e}_1, \mathbf{e}_3)$ is orthogonal to $Y = Span(\mathbf{e_2}, \mathbf{e}_4)$.

Solution

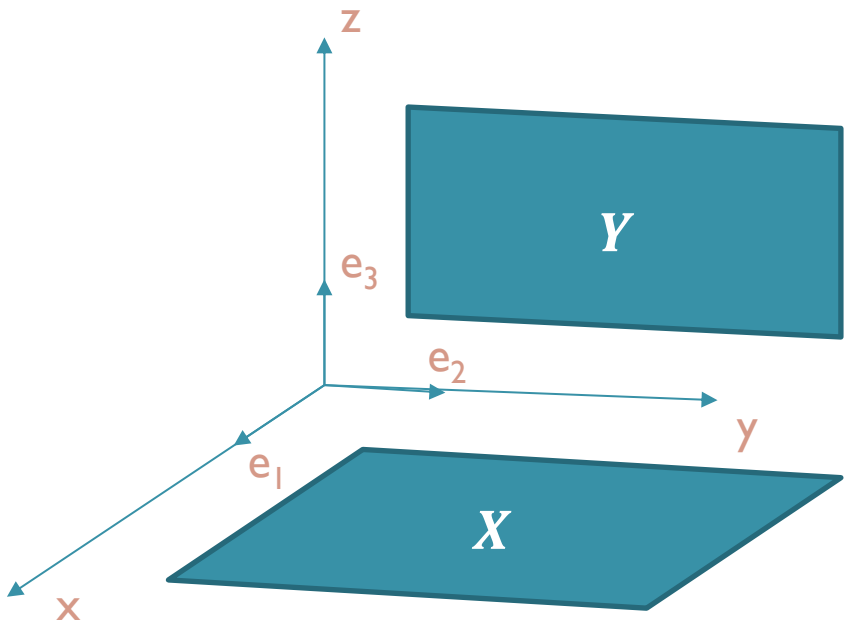
Let $\boldsymbol{x} \in X, \boldsymbol{y} \in Y$, then $\boldsymbol{x} = \begin{bmatrix} x_1 \\ 0 \\ x_3 \\ 0 \end{bmatrix}, \boldsymbol{y} = \begin{bmatrix} 0 \\ y_2 \\ 0 \\ y_4 \end{bmatrix}$.

Then $\boldsymbol{x}^T \boldsymbol{y} = x_1 \cdot 0 + 0 \cdot y_2 + x_3 \cdot 0 + 0 \cdot y_4 = 0$.

Because $\boldsymbol{x} \perp \boldsymbol{y}$ for any $\boldsymbol{x} \in X, \boldsymbol{y} \in Y$, we conclude that $X \perp Y$.

Example 2: Orthogonal or not?

In $\mathbb{R}^3$, let $X = Span(\mathbf{e}_1, \mathbf{e}_2)$, $Y = Span(\mathbf{e}_2, \mathbf{e}_3)$.



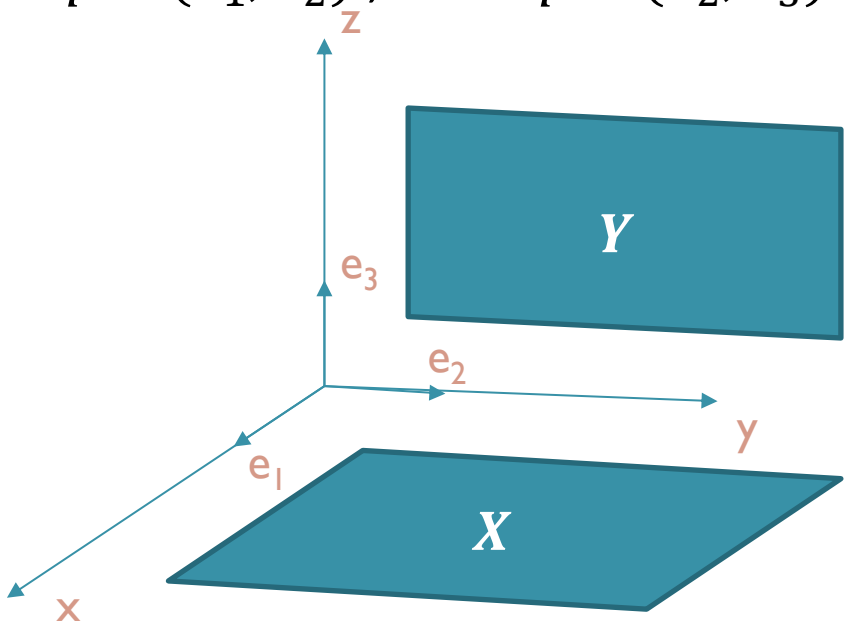
Questions

1) If $\mathbf{x} \in X$ and $\mathbf{y} \in Y$, is $\boldsymbol{x} \perp \boldsymbol{y}$ **?**

2) Is $X \perp Y$?

Example 2: Orthogonal or not?

In $\mathbb{R}^3$, let $X = Span(\mathbf{e}_1, \mathbf{e}_2)$, $Y = Span(\mathbf{e}_2, \mathbf{e}_3)$.



Questions

1) If $\mathbf{x} \in X$ and $\mathbf{y} \in Y$, is $\boldsymbol{x} \perp \boldsymbol{y}$?

2) Is $X \perp Y$?

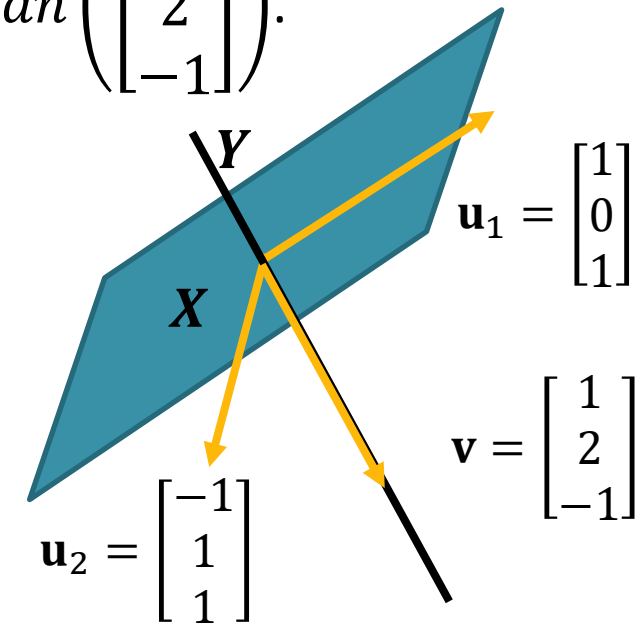
No: $\begin{bmatrix} x_1 \\ x_2 \\ 0 \end{bmatrix}^T \begin{bmatrix} 0 \\ y_2 \\ y_3 \end{bmatrix} = x_2 y_2 \neq 0$

No.

Example 3: plane $\perp$ line in $\mathbb{R}^3$

Let $X = Span\left(\begin{bmatrix}1\\0\\1\end{bmatrix}, \begin{bmatrix}-1\\1\\1\end{bmatrix}\right)$ and $Y = Span\left(\begin{bmatrix}1\\2\\-1\end{bmatrix}\right)$.

Show that $X \perp Y$.



Solution

Write $X = Span(\mathbf{u}_1, \mathbf{u_2}), Y = Span(\mathbf{v})$.

$$\mathbf{u}_1^{\mathrm{T}}\mathbf{v} = \begin{bmatrix}1 & 0 & 1\end{bmatrix}\begin{bmatrix}1\\2\\-1\end{bmatrix} = 0$$

$$\mathbf{u}_2^{\mathrm{T}}\mathbf{v} = \begin{bmatrix}-1 & 1 & 1\end{bmatrix}\begin{bmatrix}1\\2\\-1\end{bmatrix} = 0$$

For any $\mathbf{x} = \alpha_1\mathbf{u_1} + \alpha_2\mathbf{u_2} \in X$ and any $\mathbf{y} = \beta\mathbf{v} \in Y$,

$\mathbf{x}^T\mathbf{y} = (\alpha_1\mathbf{u_1} + \alpha_2\mathbf{u_2})^T(\beta\boldsymbol{v})$

$= \alpha_1\beta\mathbf{u_1^T}\mathbf{v} + \alpha_2\beta\mathbf{u_2^T}\mathbf{v} = 0$

Therefore, $X \perp Y$.

To see if $X \perp Y$, we only need to check that their basis vectors are orthogonal.

Orthogonal Complement

Definition

For a subspace Y of $\mathbb{R}^n$, the set of vectors in $\mathbb{R}^n$ that are orthogonal to every vector in Y will be denoted $Y^\perp$. Thus,

$$Y^\perp = \{\mathbf{x} \in \mathbb{R}^n | \mathbf{x}^T\mathbf{y} = 0 \quad \text{for every} \quad \mathbf{y} \in Y\}$$

The set $Y^\perp$ is called the ***orthogonal complement*** of Y

- **Examples:**
  - If $X = Span(\mathbf{e}_1)$, then $X^\perp = Span(\mathbf{e}_2, \mathbf{e}_3)$.
  - If $Y = Span(\mathbf{e}_2)$, then $Y^\perp = Span(\mathbf{e}_1, \mathbf{e}_3)$.
  - X and Y are orthogonal subspaces: $X \perp Y$. However, Y is not the orthogonal complement of X: $Y \neq X^\perp$.
- $Y^\perp$ is the "biggest" subspace orthogonal to Y.

Theorem

Theorem

If X and Y are orthogonal subspaces of $\mathbb{R}^n$, then

$$X \cap Y = \{\mathbf{0}\}$$

Proof

If $\mathbf{v} \in X \cap Y$, then $\mathbf{v} \in X$ and $\mathbf{v} \in Y$,

Since X and Y are orthogonal subspaces,

$$\mathbf{v}^T\mathbf{v} = 0$$

Thus, $\mathbf{v}$ must be zero vectors, and $X \cap Y = \{\mathbf{0}\}$.

Theorem

Theorem

If Y is a subspace of $\mathbb{R}^n$, then $Y^{\perp}$ is also a subspace of $\mathbb{R}^n$.

Proof

- $Y^{\perp}$ is nonempty as $\mathbf{0}$ is an element of $Y^{\perp}$.
- If $\mathbf{x} \in Y^{\perp}$ and α is a scalar, then for every $\mathbf{y} \in Y$

$$(\alpha\mathbf{x})^T\mathbf{y} = \alpha\mathbf{x}^T\mathbf{y} = \alpha \cdot 0$$

$\Rightarrow \alpha\mathbf{x} \in Y^{\perp}$.

- If $\mathbf{x}_1, \mathbf{x}_2$ are elements of $Y^{\perp}$, then

$$(\mathbf{x}_1 + \mathbf{x}_2)^T\mathbf{y} = \mathbf{x}_1{}^T\mathbf{y} + \mathbf{x}_2{}^T\mathbf{y} = 0 + 0 = 0$$

$\Rightarrow \mathbf{x}_1 + \mathbf{x}_2 \in Y^{\perp}$.

Therefore, $Y^{\perp}$ is a subspace of $\mathbb{R}^n$.

Fundamental subspaces

- Let A be an $m \times n$ matrix.
- As a linear transformation, $A: \mathbb{R}^n \to \mathbb{R}^m$.
  - $N(A)$ is a subspace of $\mathbb{R}^n$ (where $A\mathbf{x} = \mathbf{0}$).
  - The range of A = the column space of A (subspace of $\mathbb{R}^m$):

$$R(A) = \{\mathbf{b} \in \mathbb{R}^m | \mathbf{b} = A\mathbf{x} \text{ for some } \mathbf{x} \in \mathbb{R}^n\}$$

- As a linear transformation, $A^T: \mathbb{R}^m \to \mathbb{R}^n$
  - $N(A^T)$ is a subspace of $\mathbb{R}^m$ (where $A^T\mathbf{x} = \mathbf{0}$).
  - The column space of A^T, $R(A^T)$ is a subspace of $\mathbb{R}^n$

$$R(A^T) = \{\mathbf{y} \in \mathbb{R}^n | \mathbf{y} = A^T\mathbf{x} \text{ for some } \mathbf{x} \in \mathbb{R}^m\}$$

  - The column space of A^T is basically the row space of A, except row vectors "stands up" and become column vectors in $\mathbb{R}^n$. Thus, $\mathbf{y} \in R(A^T)$ if and only if $\mathbf{y}^T$ is in the row space of A.
- $N(A), R(A^T)$ in $\mathbb{R}^n$, $N(A^T), R(A)$ in $\mathbb{R}^m$ are called ***fundamental subspaces***.

Fundamental Subspaces Theorem

Theorem

If A is an $m \times n$ matrix, then

$$N(A) = R(A^T)^{\perp},$$
$$N(A^T) = R(A)^{\perp}.$$

Explanation

Transpose, Orthogonal, Null space

$$a_1x_1 + a_2x_2 + \cdots = 0 \Leftrightarrow \begin{bmatrix} a_1 \\ a_2 \\ \vdots \end{bmatrix} \perp \begin{bmatrix} x_1 \\ x_2 \\ \vdots \end{bmatrix}$$

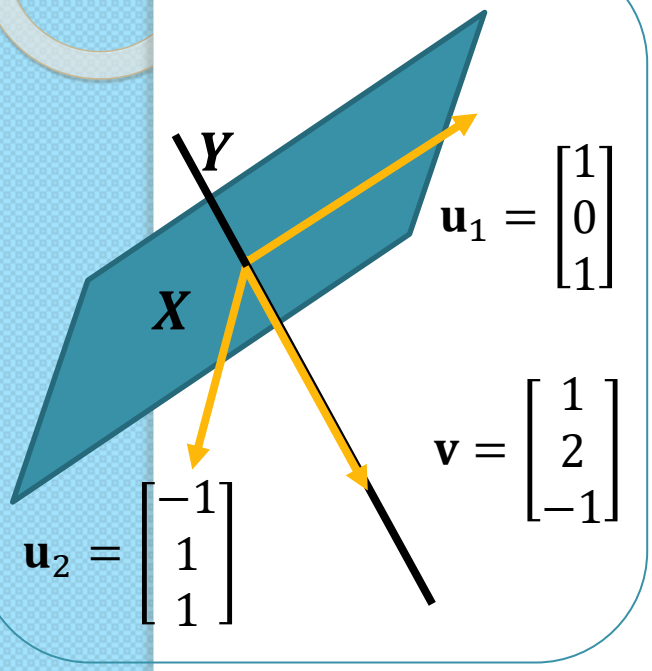
- $\mathbf{x}$ is in the null space $\leftrightarrow$ $\mathbf{x}$ satisfies m equations $a_1x_1 + a_2x_2 + \cdots = 0$

  $\leftrightarrow$ $\mathbf{x}$ is orthogonal to m column vectors $\begin{bmatrix} a_1 \\ a_2 \\ \vdots \end{bmatrix}$ (originally rows of A)

  $\leftrightarrow$ $\mathbf{x}$ is orthogonal to the span of m column vectors (originally rows of A)

  $\leftrightarrow$ $\mathbf{x}$ is orthogonal to the column space of the transpose matrix.

Example: $N(A^T) = R(A)^{\perp}$



- Let $A = \begin{bmatrix}1 & -1\\0 & 1\\1 & 1\end{bmatrix}$
- $N(A^T) = N\left(\begin{bmatrix}1 & 0 & 1\\-1 & 1 & 1\end{bmatrix}\right)$

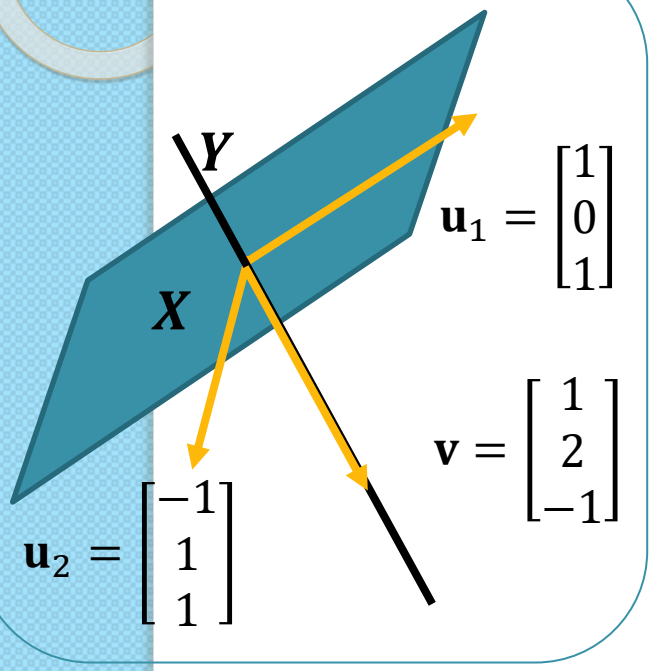
$$= \left\{\mathbf{x} = [\mathrm{a}, \mathrm{b}, \mathrm{c}]^{\mathrm{T}} \middle| \begin{matrix}1a + 0b + 1c = 0\\-1a + 1b + 1c = 0\end{matrix}\right\}$$

$$= \mathrm{Span}\left(\begin{bmatrix}1\\2\\-1\end{bmatrix}\right) = Y$$

- $R(A) = R\left(\begin{bmatrix}1 & -1\\0 & 1\\1 & 1\end{bmatrix}\right) = X$
- $Y = X^{\perp}$, i.e.

$$N(A^T) = R(A)$$

Example: $N(A) = R(A^T)^{\perp}$



- $A = [1 \quad 2 \quad -1]$
- $N(A)$
  $= N([1 \quad 2 \quad -1])$
  $= \{\mathbf{x} = [\mathrm{a}, \mathrm{b}, \mathrm{c}]^{\mathrm{T}} | a + 2b - c = 0\}$
  $= X$
- $R(A^T) = R\left(\begin{bmatrix}1\\2\\-1\end{bmatrix}\right)$
  $= Y$
- $X = Y^{\perp}$, i.e.

$$N(A) = R(A^T)^{\perp}$$

Orthogonal complement: Dimension and basis

Theorem

If S is a subspace of $\mathbb{R}^n$, then dim S + dim $S^{\perp} = n$. Furthermore, if $\{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_r\}$ is a basis for S and $\{\mathbf{x}_{r+1}, \ldots, \mathbf{x}_n\}$ is a basis for $S^{\perp}$, then $\{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_r, \mathbf{x}_{r+1}, \ldots, \mathbf{x}_n\}$ is a basis for $\mathbb{R}^n$.

Explanation

- Put the basis for S together as an $n \times r$ matrix (with rank $r = \dim S$):

$$A = [\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_r]$$

  Then $S = R(A), S^{\perp} = R(A)^{\perp} = N(A^T)$.

- Rank-nullity theorem: Because A^T is $r \times n$,

$$\text{rank}(A^T) + \dim N(A^T) = n$$
$$\Rightarrow \dim S + \dim S^{\perp} = n\,.$$

Direct Sum

Definition

If U and V are subspaces of a vector space W and each $\mathbf{w} \in W$ can be written uniquely as a sum $\mathbf{u} + \mathbf{v}$, where $\mathbf{u} \in U$ and $\mathbf{v} \in V$, then we say that W is a direct sum of U and V, and we write

$$W = U \oplus V.$$

Example: If $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is a basis of $\mathbb{R}^3$, then

$\mathbb{R}^3 = span(\mathbf{v}_1) \oplus span(\mathbf{v}_2, \mathbf{v}_3)$,

$\mathbb{R}^3 = span(\mathbf{v}_1, \mathbf{v}_2) \oplus span(\mathbf{v}_3)$.

Direct Sum Theorem

If S is a subspace of $\mathbb{R}^n$, then $\mathbb{R}^n = S \oplus S^{\perp}$.

Explanation

- Meaning: Given any subspace S, every vector $\mathbf{x}$ in $\mathbb{R}^n$ can be uniquely written as $\mathbf{x} = \mathbf{p} + \mathbf{z}$, where $\mathbf{p}$ is in S, and $\mathbf{z} = \mathbf{x} - \mathbf{p}$ is orthogonal to S.
  (Later we will call this $\mathbf{p}$ the ***projection*** of $\mathbf{x}$ onto S.)
- **Examples:**
  - In $\mathbb{R}^3$, $S = Span(\mathbf{e}_1), S^{\perp} = Span(\mathbf{e}_2, \mathbf{e}_3)$, then (uniquely) $\mathbf{x} = (x_1 e_1) + (x_2 e_2 + x_3 e_3)$.
  - If $S = Span(\mathbf{y})$, then $\mathbf{x} = \mathbf{p} + \mathbf{z}$, where $\mathbf{p} = \frac{\mathbf{x^T y}}{\mathbf{y^T y}}\mathbf{y}$ is the vector projection, and $\mathbf{z} = \mathbf{x} - \mathbf{p} \in S^{\perp}$.
  - $N(A) = R(A^T)^{\perp}$, therefore $\mathbb{R}^n = R(A^T) \oplus N(A)$.
  - $N(A^T) = R(A)^{\perp}$, therefore $\mathbb{R}^m = R(A) \oplus N(A^T)$.

→**We will use this a lot!**

Theorem: complement of complement

Theorem

If S is a subspace of $\mathbb{R}^n$, then $(S^{\perp})^{\perp} = S$.

Explanation

- Meaning: If $T = S^{\perp}$ is the orthogonal complement of S, then S is also the orthogonal complement of T.
- We simply say that S and $S^{\perp}$ are orthogonal complements (of each other).
- **Examples:**
  - In $\mathbb{R}^3$, $S = Span(\mathbf{e}_1), S^{\perp} = Span(\mathbf{e}_2, \mathbf{e}_3)$, and $(S^{\perp})^{\perp} = Span(\mathbf{e}_1) = S$.
  - In $\mathbb{R}^3$, the orthogonal complement of any plane π (with $\mathbf{0}$) is a line l (spanned by its normal vector $\boldsymbol{N}$). The orthogonal complement of line l is again the plane π.
  - $N(A) = R(A^T)^{\perp}$, therefore $N(A)^{\perp} = R(A^T)$. Similarly, $N(A^T)^{\perp} = R(A)$.

Example

Let $A = \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & 1 \\ 1 & 3 & 4 \end{bmatrix}$, find the basis for $N(A)$, $R(A^T)$, $N(A^T)$, $R(A)$.

Solution

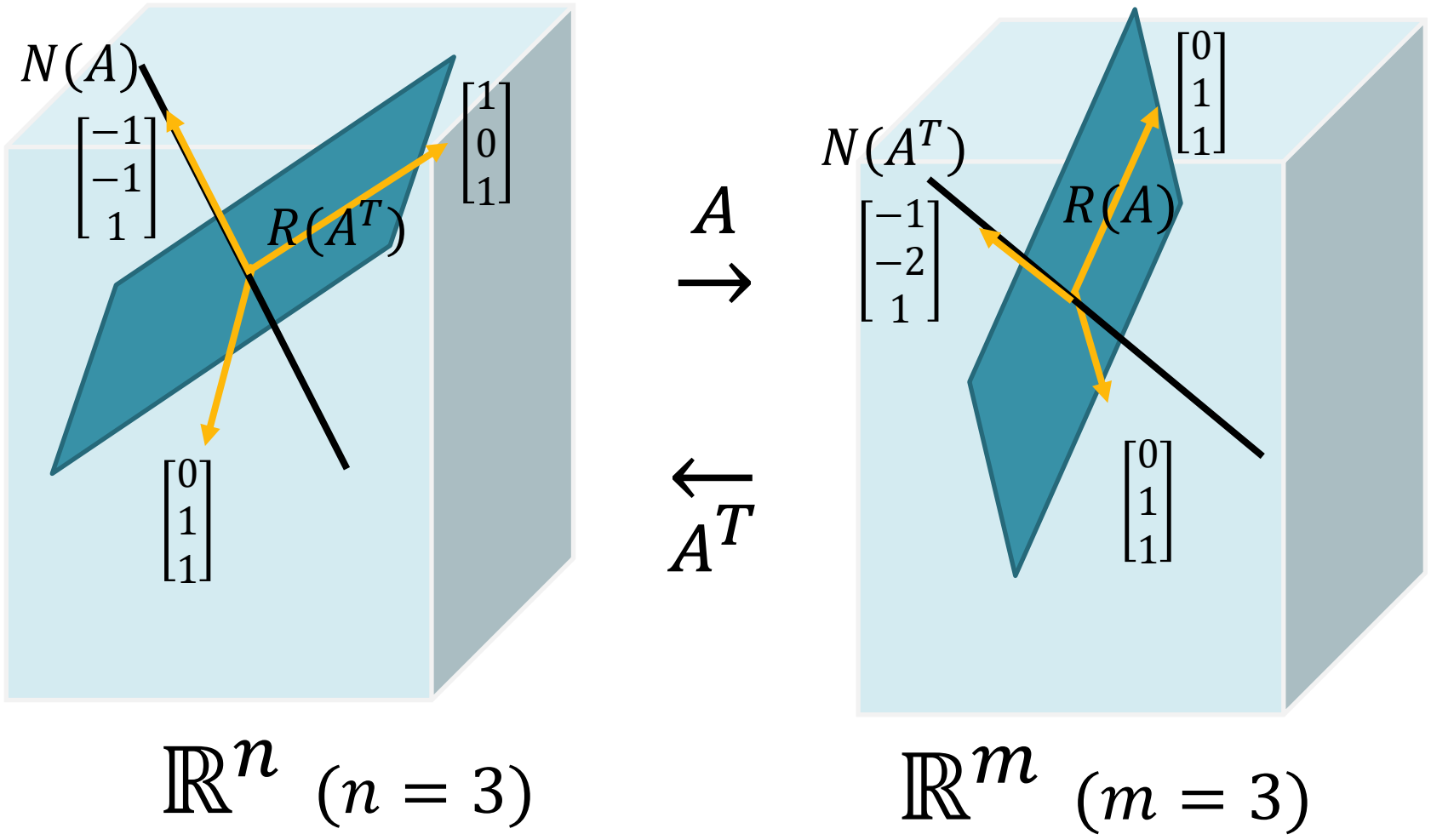
- We can find $N(A)$, $R(A^T)$ by transforming A into reduced row echelon form:

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

- $[1,0,1]$ and $[0,1,1]$ form a basis for row space of A, so $[1,0,1]^T$ and $[0,1,1]^T$ form a basis for $R(A^T)$.
- If $\mathbf{x} \in N(A)$, then $x_1 + x_3 = 0$, $x_2 + x_3 = 0$, therefore $[-1,-1,1]^T$ form a basis for $N(A)$.
- To find $N(A^T)$, $R(A)$, similarly transform A^T to reduced echelon form.
- $[1,0,1]^T$ and $[0,1,2]^T$ form a basis for $R(A)$, $[-1,-2,1]^T$ form a basis for $N(A^T)$.

See next page for a graph

Fundamental Subspaces



$\mathbb{R}^n$ $(n = 3)$

$\mathbb{R}^m$ $(m = 3)$

Summary

For a subspace S in $\mathbb{R}^n$:

- $S \cap S^{\perp} = \{\mathbf{0}\}$
- $\dim S + \dim S^{\perp} = n$
- $\mathbb{R}^n = S \oplus S^{\perp}$
- $(S^{\perp})^{\perp} = S$

For a $m \times n$ matrix A:

- Subspaces of $\mathbb{R}^n$: $N(A), R(A^T)$.
- $N(A) = R(A^T)^{\perp}$, $\mathbb{R}^n = N(A) \oplus R(A^T)$
- Subspaces of $\mathbb{R}^m$: $R(A), N(A^T)$.
- $N(A^T) = R(A)^{\perp}, \mathbb{R}^m = R(A) \oplus N(A^T)$